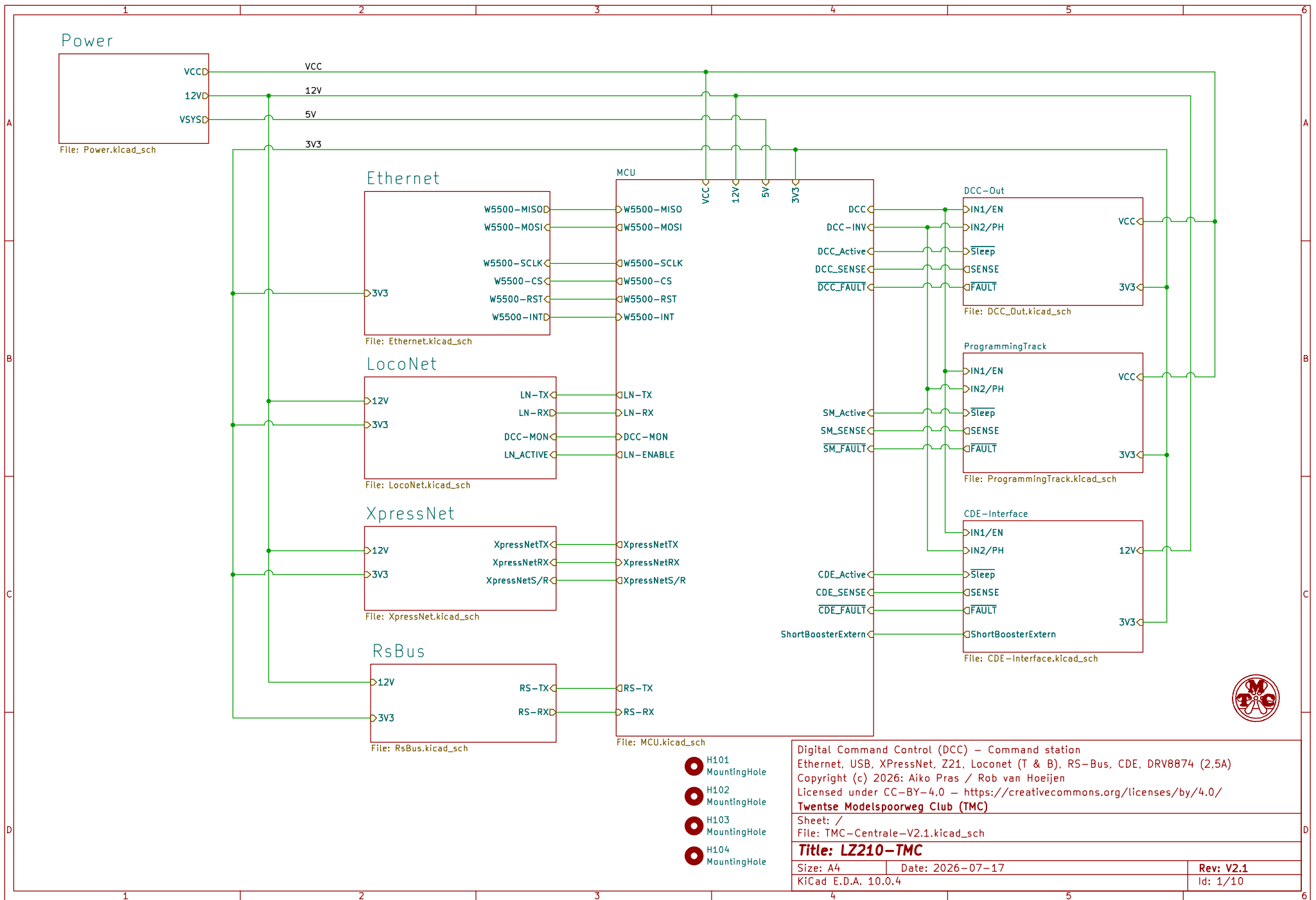
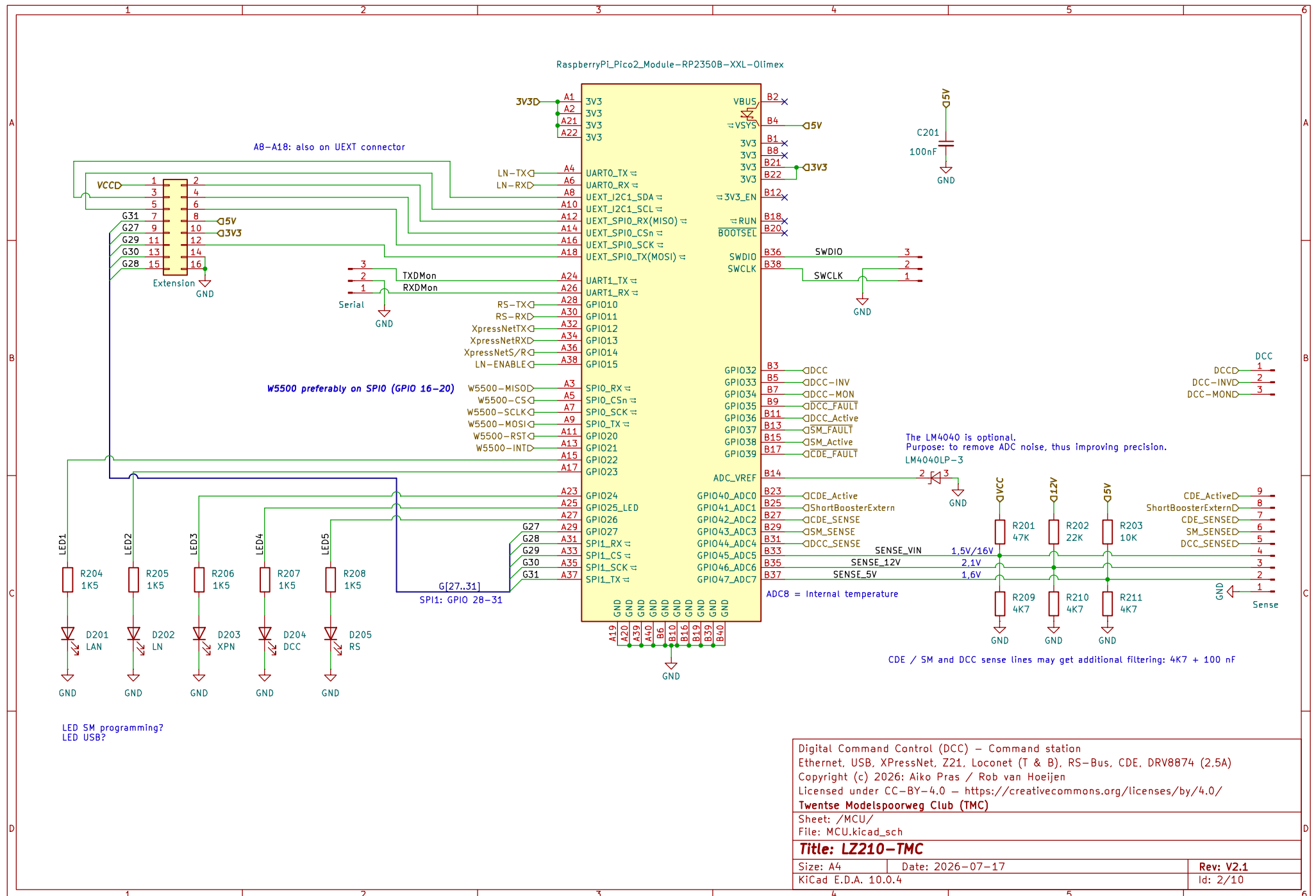
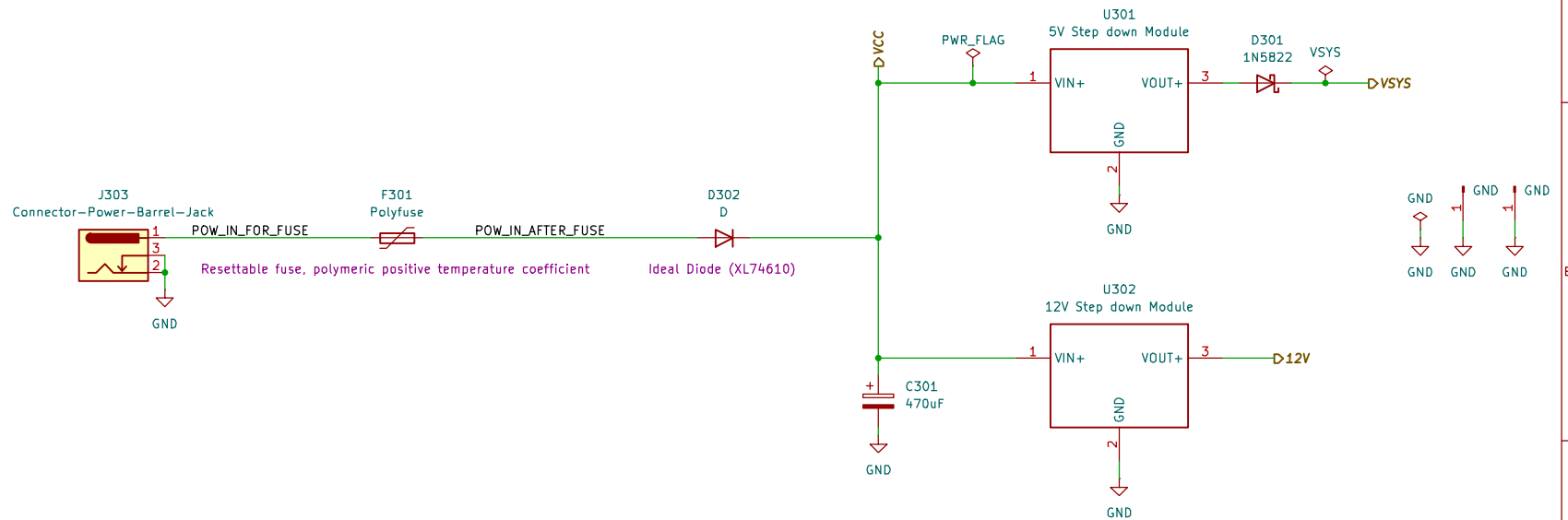








Digital Command Control (DCC) – Command station
 Ethernet, USB, XPressNet, Z21, Loconet (T & B), RS-Bus, CDE, DRV8874 (2,5A)
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Twentse Modelspoorweg Club (TMC)

Sheet: /Power/
 File: Power.kicad_sch

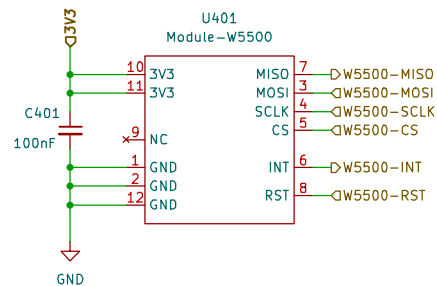
Title: LZ210-TMC

Size: A4 Date: 2026-07-17

Rev: V2.1

KiCad E.D.A. 10.0.4

Id: 3/10



Digital Command Control (DCC) – Command station
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Sheet: /Ethernet/
File: Ethernet.kicad_sch

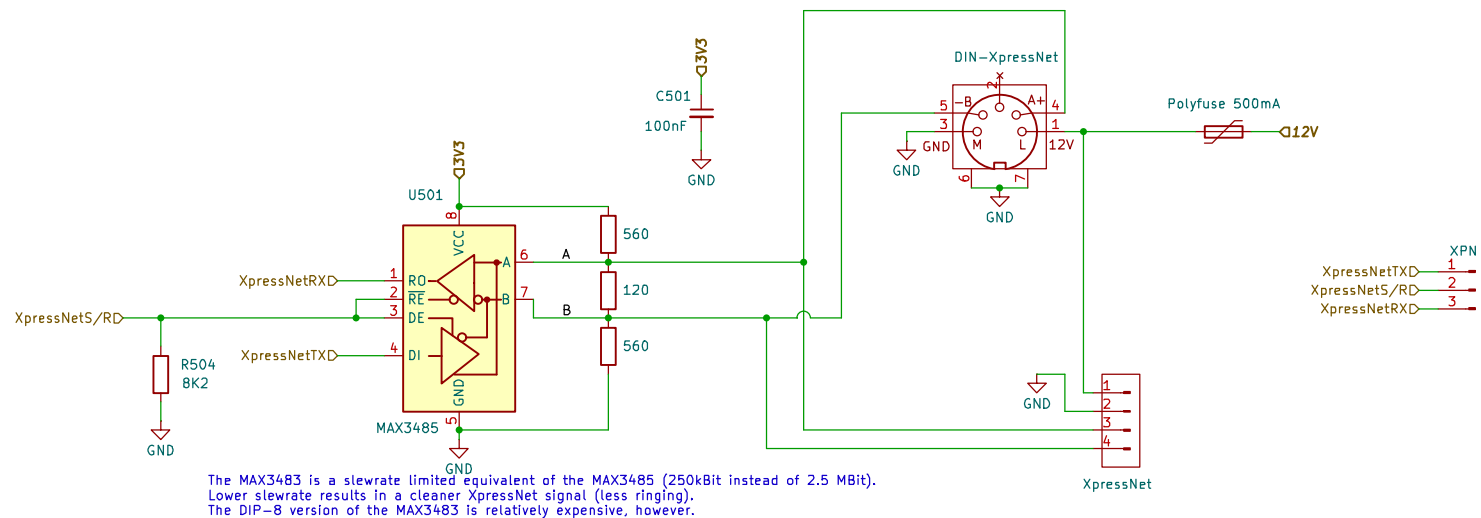
Title: LZ210-TMC

Size: A4 Date: 2026-07-17

Rev: V2.1

KiCad E.D.A. 10.0.4

Id: 4/10



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Sheet: /XpressNet/
File: XpressNet.kicad_sch

Title: LZ210-TMC

Size: A4 Date: 2026-07-17

Rev: V2.1

KiCad E.D.A. 10.0.4

Id: 5/10

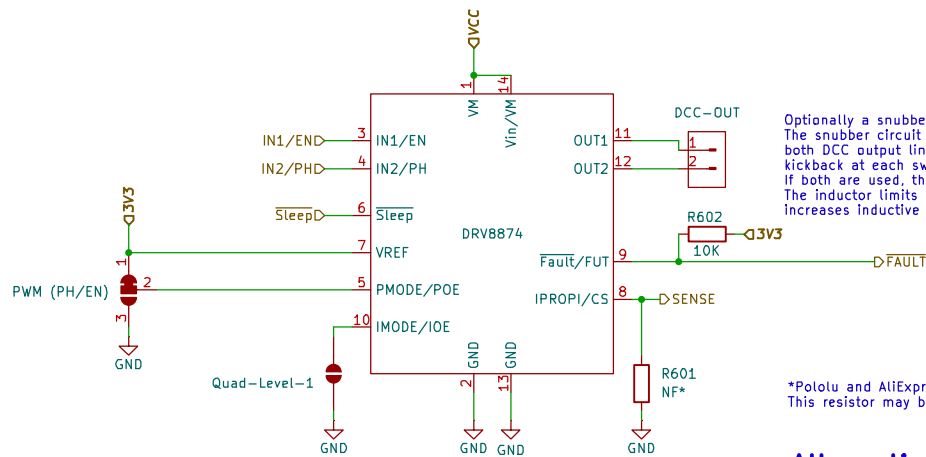
PMODE STATE	CONTROL MODE
PMODE = Logic Low	PH/EN
PMODE = Logic High	PWM
PMODE = Hi-Z	Independent Halfbridge

*DRV8871: only PWM

IMODE STATE		IMODE FUNCTION		nFAULT Response
		Current Chopping Mode	Overcurrent Response	
Quad-Level 1	RIMODE = GND	Fixed Off-Time	Automatic Retry	Overcurrent Only
Quad-Level 2	RIMODE = 20 kΩ to GND	Cycle-By-Cycle	Automatic Retry	Current Chopping and Overcurrent
Quad-Level 3	RIMODE = 62 kΩ to GND	Cycle-By-Cycle	Outputs Latched Off	Current Chopping and Overcurrent
Quad-Level 4	RIMODE = Hi-Z	Fixed Off-Time	Outputs Latched Off	Overcurrent Only

*Quad-Level 2 is chosen to facilitate inrush protection (3μs pulses)

*Pololu and AliExpress Modules include 20K SMD resistor on board (Quad-Level 2)



Optionally a snubber circuit and/or a series inductor of 4.7 to 10 μH may be connected at the DCC output connector. The snubber circuit consists of a 100 nF capacitor in series with a 10–47 Ω resistor, connected in parallel across both DCC output lines. Its primary purpose is to protect the H-bridge from voltage spikes caused by inductive kickback at each switching moment. A secondary effect is reduction of ringing on the DCC signal. If both are used, the snubber must be placed closest to the H-bridge, followed by the series inductor. The inductor limits inrush current caused by track and decoder capacitance. Note that the inductor significantly increases inductive kickback voltage – the snubber is then essential.

*Pololu and AliExpress Modules include a 2.49 KΩ on board SMD resistor between IPROPI and GND. This resistor may be replaced to obtain other Itrip / Vsense values. See table below.

Alternative 1: DRV8876

Also these modules include a 2.49 KΩ on board SMD resistor between IPROPI and GND
For the DRV8876 AIPROPI is 1000uA/A.
With Vref = 3V3, this results into an ITRIP value of 1.35A.

Alternative 2: DRV8871

*Adafruit and AliExpress DRV8871 Modules include a 30 KΩ on board SMD resistor between ILIM and GND
This gives a current limit of approximately 2A
DRV8871 modules have a different (larger) footprint, and do not support a nFault output.

Formula: $ITRIP (A) \times AIPROPI (\mu A/A) = VREF (V) / RIPROPI (\Omega)$
DRV8874: AIPROPI = 450 μA/A
DRV8876: AIPROPI = 1000 μA/A

Chip (Vref = 3V3)	Rsense	Itrip	Vsense – 60mA ACK	Remarks
DRV8874	2.49 kΩ	2945 mA	67 mV	Fitted on module
DRV8874	6.8 kΩ	1078 mA	184 mV	
DRV8874	15 kΩ	489 mA	405 mV	
DRV8874	27 kΩ	272 mA	729 mV	
DRV8876	2.49 kΩ	1325 mA	149 mV	Fitted on module
DRV8876	6.8 kΩ	485 mA	408mV	
DRV8876	15 kΩ	220 mA	900 mV	

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Sheet: /DCC-Out/
File: DCC_Out.kicad_sch

Title: LZ210-TMC

Size: A4 Date: 2026-07-17

Rev: V2.1

KiCad E.D.A. 10.0.4

Id: 6/10

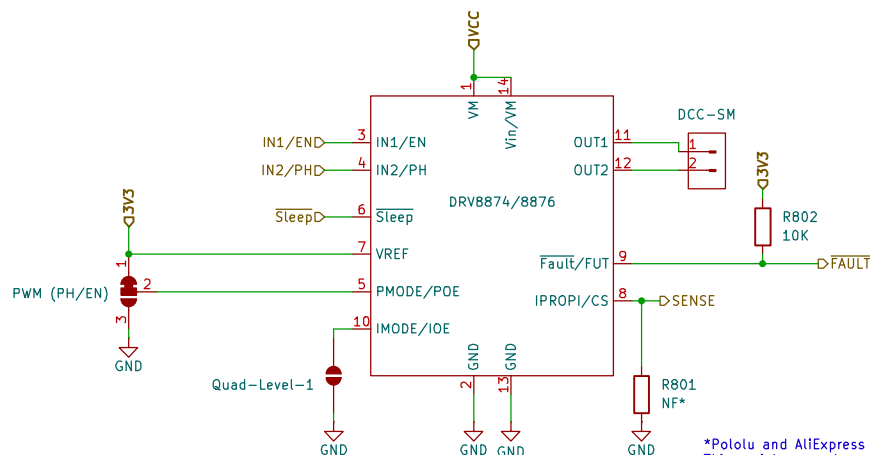
PMODE STATE	CONTROL MODE
PMODE = Logic Low	PH/EN
PMODE = Logic High	PWM
PMODE = Hi-Z	Independent Halfbridge

*DRV8871: only PWM

IMODE STATE		IMODE FUNCTION		nFAULT Response
Quad-Level 1	RIMODE = GND	Current Chopping Mode	Overcurrent Response	
Quad-Level 2	RIMODE = 20 kΩ to GND	Fixed Off-Time	Automatic Retry	Overcurrent Only
Quad-Level 3	RIMODE = 62 kΩ to GND	Cycle-By-Cycle	Automatic Retry	Current Chopping and Overcurrent
Quad-Level 4	RIMODE = Hi-Z	Cycle-By-Cycle	Outputs Latched Off	Current Chopping and Overcurrent
		Fixed Off-Time	Outputs Latched Off	Overcurrent Only

*Quad-Level 2 is chosen to facilitate inrush protection (3μs pulses)

*Pololu and AliExpress Modules include 20K SMD resistor on board (Quad-Level 2)



*Pololu and AliExpress Modules include a 2.49 KΩ on board SMD resistor between IPROPI and GND. This resistor may be replaced to obtain other Itrip / Vsense values. See table below.

Formula: $I_{TRIP} (A) \times AIPROPI (\mu A/A) = V_{REF} (V) / R_{IPROPI} (\Omega)$
 DRV8874: AIPROPI = 450 μA/A
 DRV8876: AIPROPI = 1000 μA/A

Chip (Vref = 3V3)	Rsense	Itrip	Vsense – 60mA ACK	Remarks
DRV8874	2.49 kΩ	2945 mA	67 mV	Fitted on module
DRV8874	6.8 kΩ	1078 mA	184 mV	
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DRV8876	2.49 kΩ	1325 mA	149 mV	Fitted on module
DRV8876	6.8 kΩ	485 mA	408mV	
DRV8876	15 kΩ	220 mA	900 mV	

According to RCN216, the max. current should be limited to 250 ma (+/- 20%)

Digital Command Control (DCC) – Command station
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Sheet: /ProgrammingTrack/
 File: ProgrammingTrack.kicad_sch

Title: LZ210-TMC

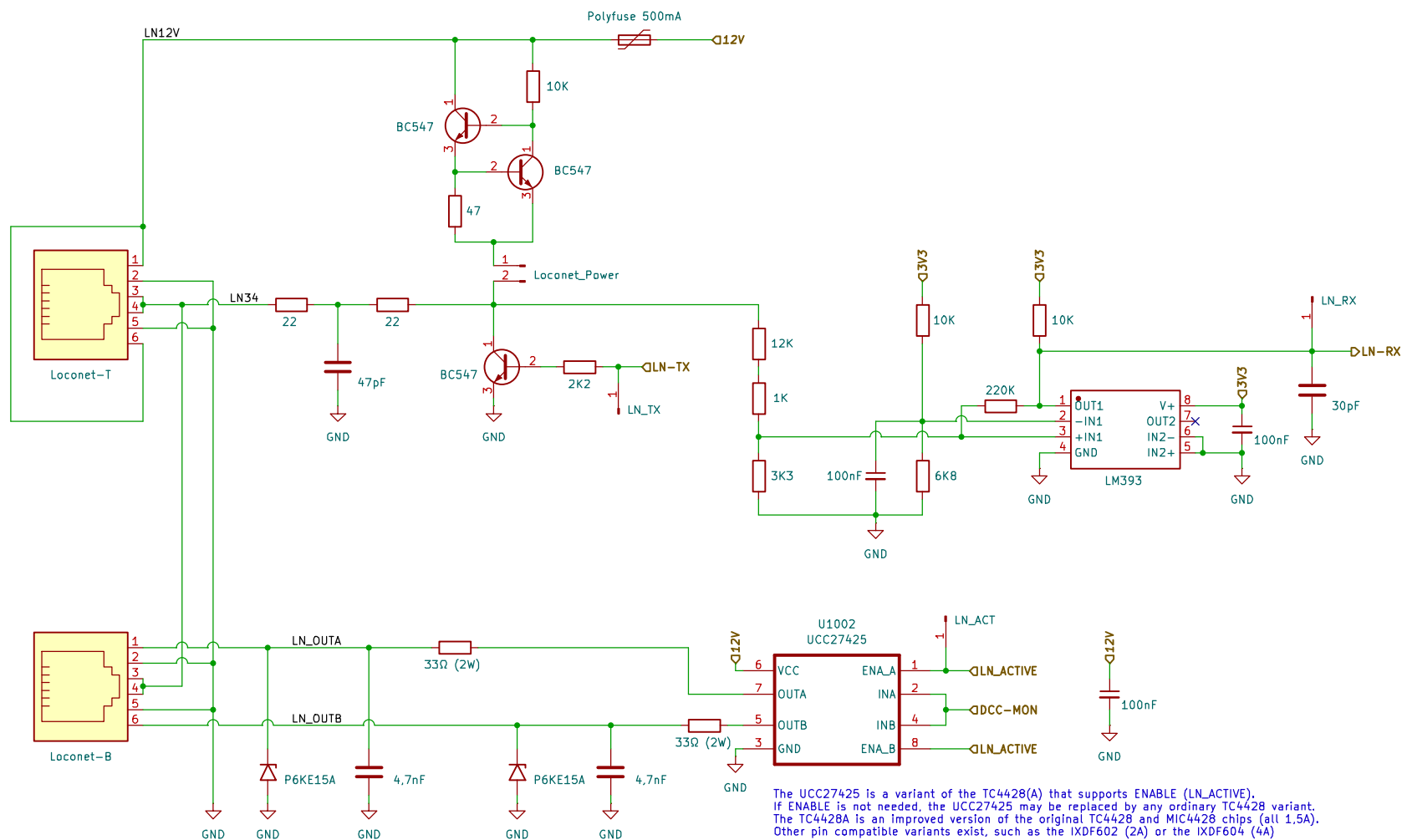
Size: A4 Date: 2026-07-17

Rev: V2.1

KiCad E.D.A. 10.0.4

Id: 8/10





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Sheet: /LocoNet/
 File: LocoNet.kicad_sch

Title: LZ210-TMC

Size: A4 Date: 2026-07-17

Rev: V2.1

KiCad E.D.A. 10.0.4

Id: 10/10